

$^{35}\text{Cl}(^3\text{He},\alpha)$ 1974Va25

$^{35}\text{Cl} \rightarrow \text{In} + ^{34}\text{Cl}$ from $J^\pi = 3/2^+$ ^{35}Cl ground state.

1974Va25: a 15-MeV ^3He beam was produced from the Utrecht 6-MV tandem accelerator. Targets were BaCl_2 99.3% enriched in ^{35}Cl on C backing. α particles were detected using seven position-sensitive detectors with FWHM=15 keV in the focal plane of the split-pole magnetic spectrograph. Measured $\sigma(E_\alpha, \theta)$. Deduced levels, J, π , L-transfers, and spectroscopic factors from DWBA analysis. Compared with mirror ^{34}S and discussed isospins.

 ^{34}Cl Levels

<u>E(level)[†]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>	<u>E(level)[†]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>	<u>E(level)[†]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>
0.0	2	0.44	3128	2	0.31	4206 10	0+2	0.20+0.05
146.36	2	1.5	3333	2	0.33	4321 10	0+2	0.007+0.04
461.00	0+2	0.09+0.26	3383	0+2	0.27+0.23	4352.7		
665.55	0+2	0.03+0.31	3545.2	1	0.007	4608	1+3	0.007+0.04
1230.24	0+2	0.23+0.07	3632.5	3	0.030	4639		
1887.5	0+2	0.06+0.14	3847 10			4715 10	2	0.20
2158.3	0+2	0.10+0.18	3937 10	0+2	0.003+0.07	4939 11	2	0.13
2180.7	2	0.07	3964 10	2	0.06	4958 11	2	0.30
2580.6	0	0.06	3982.1	3	0.040	4971 11	0+2	0.04+0.28
2611.2	2	0.23	4075 10	0+2	0.007+0.08	4998 12	0+2	0.08+0.56
2718 10	2	0.07	4075.4			5010 13	2	0.85
2721.7			4136.6	0+2	0.011+0.04			

[†] E(level) values with uncertainties are measured by 1974Va25; E(level) values without uncertainties are quoted from 1973EnVA, which are also observed by 1974Va25 but without reporting their own values.

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in 1974Va25. Spectroscopic factor $C^2S = \sigma(\theta)_{\text{exp}} / \sigma(\theta)_{\text{DWBA}} / N$, where $N=23.5$ is a normalization factor.